

Oct 7, 2025

$X_1, X_2, \dots, X_n$ are discrete random variable
and $g: \mathbb{R}^n \rightarrow \mathbb{R}$ is a function,

pmf of $Y = g(X_1, X_2, \dots, X_n)$

$$P(Y=y) = P_Y(y) = \sum_{\{(x_1, x_2, \dots, x_n) | g(x_1, x_2, \dots, x_n) = y\}} P_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n)$$

Sum of Independent Random Variables - Convolution

X, Y are independent Random Variables pmf of $X+Y$

$$Z = X+Y$$

$$P_Z(z) = P(Z=z) = P(X+Y=z)$$

$$= \sum_{\{(x,y) | x+y=z\}} P(X=x, Y=y) = \sum_x P(X=x, Y=z-x) = \sum_x P_X(x) P_Y(z-x)$$

$$= \sum_x P_X(x) P_Y(z-x)$$

Exercise: derive convolution sum by substitution law

$$P_Z(z) = \sum_x P_{Z|X}(z|x) P_X(x) = \sum_x P(Z=z | X=x) P(X=x)$$

$$= \sum_x P(X+Y=z | X=x) P(X=x)$$

$$= \sum_x P(x+Y=z | X=x) P(X=x)$$

$$= \sum_x P(Y=z-x) P(X=x)$$

$$= \sum_x P_X(x) P_Y(z-x)$$

Exercise: X, Y are discrete r.v. on $(\Omega, \mathcal{F}, P)$, and are independent

$$Z = XY$$

$$\begin{aligned} P_Z(z) &= P(Z=z) = \sum_x P(Z=z|X=x)P(X=x) \\ &= \sum_x P(XY=z|X=x)P(X=x) \\ &= \sum_x P(X \cdot Y=z|X=x)P(X=x) \\ &= \sum_{x \neq 0} P(Y=\frac{z}{x})P(X=x) + P(0 \cdot Y=z)P(X=0) \\ &= \begin{cases} \sum_{x \neq 0} P_Y(\frac{z}{x})P_X(x) + P(X=0) & z=0 \\ \sum_{x \neq 0} P_Y(z/x)P_X(x) & z \neq 0 \end{cases} \end{aligned}$$

Max and Min Problems: independent random variables $X_1, X_2, \dots, X_n$

cdf of $Y = \max(X_1, X_2, \dots, X_n)$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(\max(X_1, X_2, \dots, X_n) \leq y) \\ &= P(X_1 \leq y, X_2 \leq y, \dots, X_n \leq y) = P(X_1 \leq y)P(X_2 \leq y) \dots P(X_n \leq y) \\ &\text{if } X_i \text{'s are i.i.d.} \quad = F_X(y)F_X(y) \dots F_X(y) \end{aligned}$$

$$F_Y(y) = F_X(y)^n$$

cdf of $Z = \min(X_1, X_2, \dots, X_n)$

$$F_Z(z) = P(Z \leq z) = P(\min(X_1, X_2, \dots, X_n) \leq z)$$

$$= 1 - P(\min(X_1, X_2, \dots, X_n) > z)$$

$$= 1 - P(X_1 > z, X_2 > z, \dots, X_n > z)$$

$$= 1 - P(X_1 > z) P(X_2 > z) P(X_3 > z) \dots P(X_n > z)$$

$$= 1 - [1 - P(X_1 \leq z)] [1 - P(X_2 \leq z)] \dots [1 - P(X_n \leq z)]$$

$$= 1 - (1 - F_X(z))(1 - F_X(z)) \dots (1 - F_X(z))$$

if X_i are i.i.d

$$F_Z(z) = 1 - (1 - F_X(z))^n$$

Moment of Discrete Random Variables

Expectation $E(X) = \sum_x x P_X(x)$

Law of the unconscious Statistician (LOTUS)

for any $g: \mathbb{R} \rightarrow \mathbb{R}$, and $X, Y = g(X)$ is a R.V.

$$E(Y) = E(g(X)) = \sum_x g(x) P_X(x)$$

proof of LOTUS

$$P_Y(y) = P(Y=y) = \sum_{\{x | g(x)=y\}} P(X=x)$$

$$= \sum_{x | g(x)=y} P_X(x)$$

$$\therefore E(Y) = \sum_y y P_Y(y) = \sum_y y \sum_{x | g(x)=y} P_X(x)$$

$$= \sum_{x | g(x)=y} g(x) P_X(x) = \sum_x g(x) P_X(x)$$

Properties of Expectation

a) If $X \geq 0$ a.s. $\Rightarrow E(X) \geq 0$

b) If $X = c$ a.s. for $c \in \mathbb{R} \Rightarrow E(X) = c$

c) $a, b \in \mathbb{R}$, $E(aX + bY) = aE(X) + bE(Y)$

d) If $X: \Omega \rightarrow \mathbb{Z}^{\geq 0}$

$$E(X) = \sum_{n \geq 0} P(X > n)$$

e) X, Y are independent

$$E(XY) = E(X)E(Y)$$

Relation between probability and expectation

$$I_B(x) = \begin{cases} 0 & x \notin B \\ 1 & x \in B \end{cases}$$

$$Y = I_B(X)$$

$$E(Y) = E(I_B(X)) = \sum_x I_B(x) P(x)$$

$$= \sum_{x \in B} P(x) = P(X \in B) = P_X(B) = P(\{\omega \mid X(\omega) \in B\})$$